

ERZIN, Russian Federation

WMO: 363070

Lat: 50.258N

Lon: 95.171E

Elev: 1101

StdP: 88.78

Time Zone: 7.00 (E07)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -41.8	(c) -39.9	(d) -44.8	(e) 0.1	(f) -41.6	(g) -42.9	(h) 0.1	(i) -39.7	(j) 3.3	(k) -27.3	(l) 2.8	(m) -28.4	(n) 1.0	(o) 110	(p) 0.613

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.0	(c) 30.9	(d) 14.9	(e) 29.1	(f) 14.5	(g) 27.3	(h) 14.1	(i) 16.7	(j) 26.0	(k) 16.0	(l) 24.8	(m) 15.3	(n) 23.9	(o) 2.1	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.1	11.5	18.5	13.1	10.7	18.3	11.9	9.9	18.5	51.1	25.7	48.9	24.7	46.7	24.3	20.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.1	(b) 5.1	(c) 4.4	(e) -43.2	(f) 34.6	(g) 2.9	(h) 1.5	(i) -45.3	(j) 35.6	(k) -47.0	(l) 36.5	(m) -48.6	(n) 37.3	(o) -50.7	(p) 38.4		
(a) 6.1	(b) 5.1	(c) 4.4	(e) -43.3	(f) 18.5	(g) 2.8	(h) 1.3	(i) -45.3	(j) 19.4	(k) -47.0	(l) 20.1	(m) -48.6	(n) 20.8	(o) -50.6	(p) 21.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.3	-31.5	-27.5	-14.6	3.3	10.5	18.0	19.7	17.5	10.9	2.4	-12.2	-25.3
	DBStd	18.83	4.10	5.60	7.05	6.67	4.91	3.94	3.25	3.59	4.74	5.27	6.11	5.20
	HDD10.0	5416	1288	1049	762	211	56	2	0	1	46	239	667	1096
	HDD18.3	7674	1546	1283	1021	450	246	53	23	59	227	494	917	1354
	CDD10.0	929	0	0	0	11	71	241	299	233	71	3	0	0
	CDD18.3	146	0	0	0	0	3	42	64	33	3	0	0	0
	CDH23.3	1719	0	0	0	2	71	526	678	391	53	0	0	0
	CDH26.7	509	0	0	0	0	12	161	212	118	7	0	0	0
Wind	WSAvg	1.9	1.0	1.1	1.4	2.4	2.8	2.6	2.3	2.0	2.2	2.1	1.7	1.3
Precipitation	PrecAvg	193	5	5	6	7	12	25	47	41	17	8	11	9
	PrecMax	281	15	12	16	21	26	58	125	119	39	27	25	17
	PrecMin	119	1	0	1	0	1	5	12	5	3	0	4	1
	PrecStd	49	3	3	4	6	7	11	26	27	11	8	6	4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-19.0	-9.3	7.1	22.5	28.9	32.9	33.5	32.9	27.7	18.9	5.7	-9.5
		MCWB	-19.7	-10.9	1.6	9.5	11.9	16.5	16.4	15.3	13.0	7.8	-0.2	-10.5
	2%	DB	-21.5	-13.1	2.4	19.5	25.5	30.3	31.1	29.7	24.8	16.1	1.9	-13.0
		MCWB	-22.0	-14.1	-0.7	8.2	11.2	14.3	15.6	14.6	12.2	6.5	-1.8	-13.8
	5%	DB	-23.4	-16.2	-1.6	16.3	22.8	28.5	28.9	27.5	22.1	13.1	-0.9	-15.8
		MCWB	-23.8	-16.9	-3.4	6.6	10.2	13.9	15.4	14.4	11.1	5.0	-3.1	-16.3
	10%	DB	-25.3	-19.0	-4.4	13.5	20.5	26.5	27.2	25.2	19.5	10.4	-3.3	-17.8
		MCWB	-25.6	-19.6	-5.7	5.5	8.9	13.3	14.7	13.8	10.0	3.7	-5.2	-18.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-19.5	-10.7	2.3	10.3	13.2	17.3	18.4	17.1	14.4	8.1	0.3	-10.3
		MCDB	-18.9	-9.2	6.7	21.4	25.4	29.2	26.8	25.8	24.0	17.9	4.9	-9.7
	2%	WB	-22.0	-14.1	-0.8	8.6	11.6	16.0	17.1	16.3	13.0	6.6	-1.5	-13.8
		MCDB	-21.5	-13.1	2.3	18.2	24.2	25.9	26.6	24.9	23.3	15.2	1.3	-13.1
	5%	WB	-23.8	-16.9	-3.2	7.1	10.6	15.2	16.3	15.5	11.8	5.3	-3.0	-16.3
		MCDB	-23.4	-16.2	-1.6	15.4	21.9	24.4	25.0	24.2	20.9	12.2	-0.6	-15.9
	10%	WB	-25.6	-19.6	-5.6	5.7	9.2	14.4	15.6	14.6	10.6	4.0	-5.1	-18.1
		MCDB	-25.3	-19.1	-4.4	12.8	19.5	23.8	24.0	23.4	18.5	9.9	-3.2	-17.8
Mean Daily Temperature Range	5% DB	MDBR	9.2	11.3	12.4	11.3	13.0	13.0	12.0	12.4	12.9	11.5	9.1	8.2
		MCDBR	9.7	12.0	11.8	15.8	16.5	15.8	15.0	15.8	15.8	14.8	11.4	8.1
		MCWBR	9.2	11.1	8.7	7.9	7.6	5.6	5.6	6.1	7.0	7.8	8.1	7.6
	5% WB	MCDBR	9.7	12.0	11.2	14.3	15.4	12.7	12.1	12.7	14.5	13.8	11.0	7.9
		MCWBR	9.2	11.0	8.2	7.5	7.6	5.0	5.3	5.6	6.8	7.6	7.8	7.5
Clear-Sky Solar Irradiance	taub	0.242	0.269	0.311	0.351	0.365	0.359	0.362	0.343	0.306	0.290	0.270	0.245	
	taud	2.256	2.209	2.205	2.275	2.268	2.378	2.396	2.421	2.477	2.446	2.315	2.226	
	Ebn at Noon	811	871	890	891	891	898	889	891	895	845	763	739	
	Edh at Noon	68	94	115	121	127	115	112	103	87	74	65	60	
All-Sky Solar Radiation	RadAvg	1.32	2.30	3.91	5.35	6.40	6.75	6.25	5.42	4.29	2.70	1.41	1.02	
	RadStd	0.06	0.11	0.14	0.27	0.27	0.31	0.36	0.30	0.30	0.17	0.10	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (5 neighbors)	N/A	N/A	-2.23	N/A	-0.88	N/A	N/A	N/A	N/A		

Nomenclature: See separate page